"SmartStudy": A Machine Learning Approach to Optimizing Student Study Habits

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Abstract

Students frequently struggle to balance their extracurricular and academic obligations. Although there exist online study tools and general recommendations for these students, many of these resources lack personalization. The machine learning model presented in this research is intended to suggest individualized academic habits, including the optimal number of study hours per week, the recommended hours of tutoring, recommended lecture attendance, as well as a recommendation for involvement in sports, music, volunteering and extracurricular activities (measured as a boolean). The Students Performance Dataset [1] is a dataset that offers exactly this type of data on 2392 high school students labeled with their resulting grades, and was therefore used to train the model. A random forest regression model was used as a baseline, paired with a coordinate descent algorithm for optimizing the study habits (as features of the input data) with respect to predicted GPA. Various iterations of this model were compared using mean squared error and mean absolute error on a 70/15/15 testing split. The most successful iteration of the baseline was with TabPFN and Bayesian optimization. The final SmartStudy package also includes a k-nearest neighbors search for similar students, to offer inspiration to users. To improve the delivery of recommendations, GPT-4 was used to interpret the model's output to the student. The GitHub repo with the source code, dataset, and demo can be accessed here.

1 Introduction

Students often feel overwhelmed by the demands of coursework, deadlines, and extracurricular commitments, making it difficult to establish effective study habits. Adding to these challenges, research shows that factors from personal and family background, such as parental socioeconomic status, can significantly affect academic achievement. To address these issues, we developed a comprehensive pipeline tool that leverages machine learning to support student success. The pipeline consists of three key components: a predictive model that estimates a student's academic performance, a secondary model that recommends personalized lifestyle and habit optimizations (e.g., ideal weekly study time, tutoring needs, or lecture attendance), and a final stage that uses GPT to translate these optimizations into actionable, text-based advice. This integrated system is designed to enhance students' time management, organizational skills, and overall academic outcomes.

2 Related Work

The application of machine learning to develop personalized study recommendations based on students' backgrounds and learning behaviour is an area of active research.

In a randomized controlled trial by Agrawal et al., the impact of personalized content recommendations within an educational application for children was assessed [2]. Their findings indicated a 60% increase in the consumption of recommended content and a 14% overall increase in app usage, highlighting the efficacy of personalized recommendations in enhancing student engagement.

In the context of higher education, Jiang et al. developed a goal-based course recommendation system utilizing recurrent neural networks [3]. This system provided personalized course suggestions aligned with students' academic goals and prior knowledge, demonstrating potential in improving course selection processes and academic outcomes.

Another study by Delianidi et al. introduced DK-PRACTICE, an intelligent educational platform that leverages machine learning to offer personalized learning content recommendations based on students' knowledge states [4]. By analyzing individual learning behaviors, the platform delivers tailored educational materials to address specific knowledge gaps, promoting adaptive learning experiences.

While these studies focused on personalization through feedback, Taylor et al. proposed an integrated machine learning and simulation framework designed to enhance personalized learning systems [5]. Their methodology aimed to adapt educational tasks to individual student needs, thereby improving academic performance.

These studies collectively highlight the potential of machine learning in creating personalized educational tools that consider individual student backgrounds and behaviors. Our research builds upon this foundation by developing a machine learning model that recommends personalized lifestyle changes such as optimal weekly study time, tutoring needs, and lecture attendance to enhance students' academic success.

3 Problem Formulation

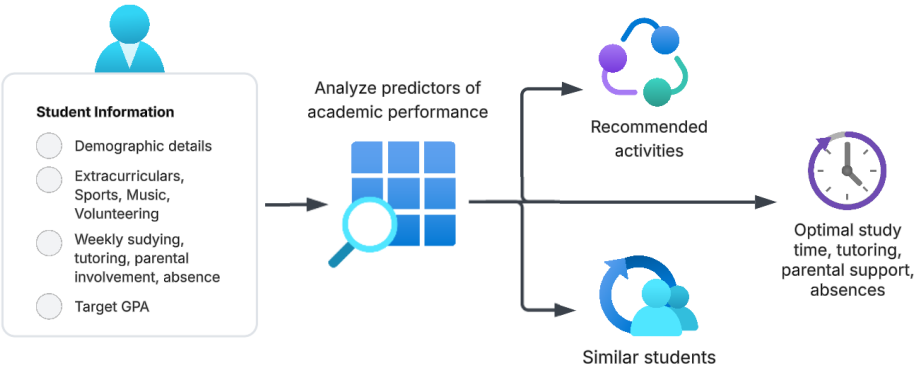


Figure 1: Task schematic for SmartStudy.

SmartStudy aims to help students navigate academics with the rest of their lifestyle by providing actionable and personalized recommendations. The task schematic in Figure 1 demonstrates how the user will first provide their demographic details, extracurricular activities, study habits, and their desired grade. Then, the model should output changes to the student's details to optimize their academic performance. The final output will be a recommended number of hours for weekly studying, level of parental support, and recommendations for whether the student should have tutoring, sports, music, volunteering, and extracurriculars. Although this is not necessary, the model can also output some students that are similar to the user, which can serve as inspiration.

4 Data and Preprocessing

This study utilized the Students Performance Dataset from Kaggle [1], which includes 2,392 records of high school students. The dataset contains demographic information, study habits, parental involvement, extracurricular activities, and academic performance metrics, making it suitable for development of a personalized recommendation system for students.

The dataset required minimal preprocessing, as it contained no missing or duplicate entries. All features were numeric, with categorical variables such as Ethnicity, ParentalEducation, ParentalSupport, and GradeClass already one-hot encoded.

To ensure equal feature importance and enhance model stability, we standardized features using StandardScaler from scikit-learn [6] to preserve the original distribution shape and the relative relationships between features. Standardization transforms each feature using the equation below, where x represents the original feature value, μ is the mean of the feature, σ is the standard deviation, and z is the standardized feature value.

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

4.1 Exploratory Data Analysis

To better understand the relationship between student attributes and academic performance, we conducted an exploratory data analysis (EDA) summarized by the figures below.

4.1.1 Correlation Heatmap

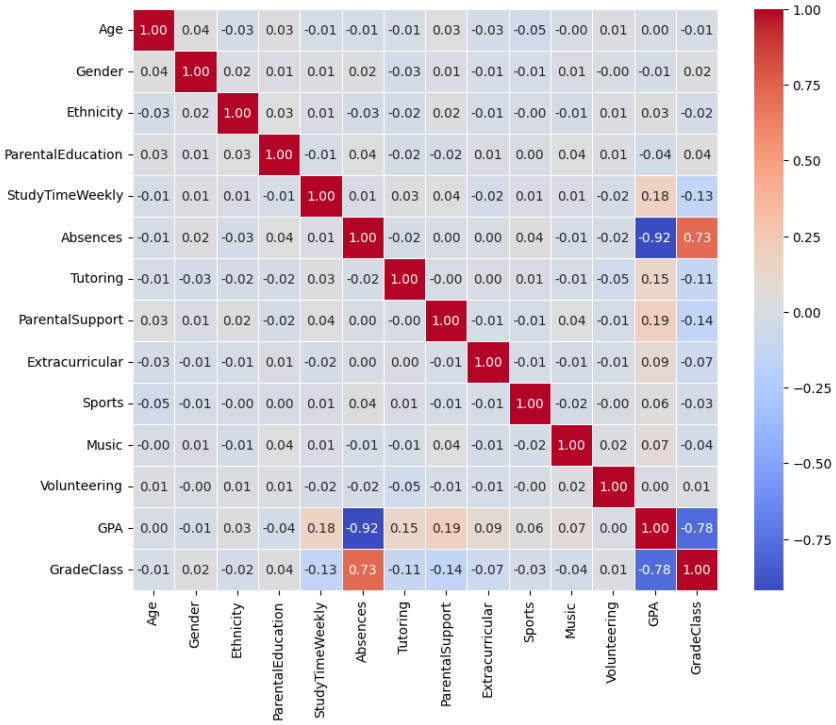


Figure 2: Correlation heatmap between student features and GPA.

Figure 2 highlights the most significant relationships using Pearson correlation, showing a strong negative correlation between absences and GPA (-0.92) and between grade class and GPA (-0.78). Weekly study time shows a moderate positive correlation with GPA (0.18), while tutoring and parental support exhibit weaker positive correlations (0.15 and 0.19, respectively). Extracurricular involvement and weekly study time exhibit minimal impact. Volunteering students appear to have slightly lower GPA than those involved in other activities. These results support the importance of consistent attendance and supplementary academic resources.

4.2 Study Time and GPA Trends

Figure 3 illustrates weak positive trends between weekly study time and GPA. The presence of parental support correlates with slightly higher GPA, regardless of parental education level.

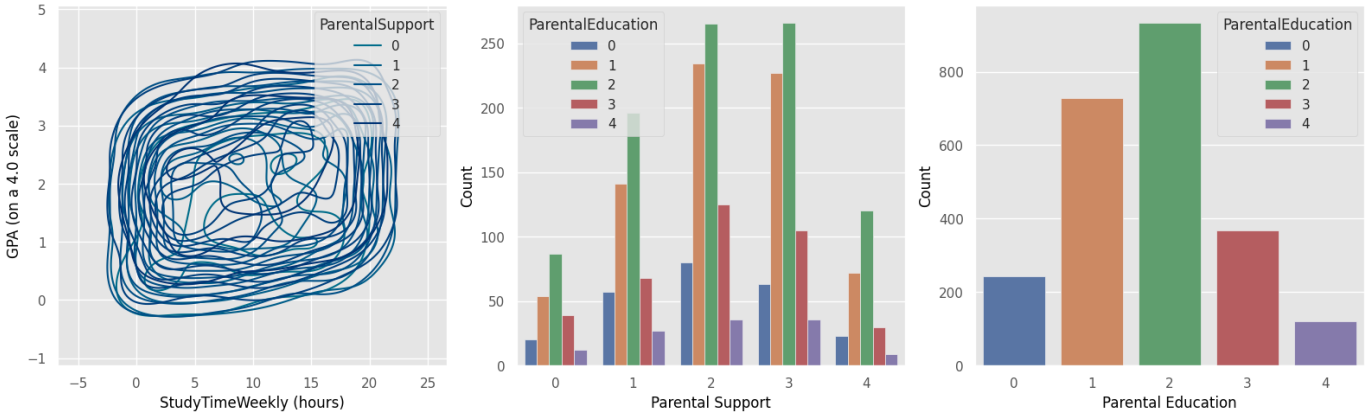


Figure 3: Relationship between weekly study time, GPA, and parental support.

4.2.1 Tutoring, Support, and Study Time Patterns

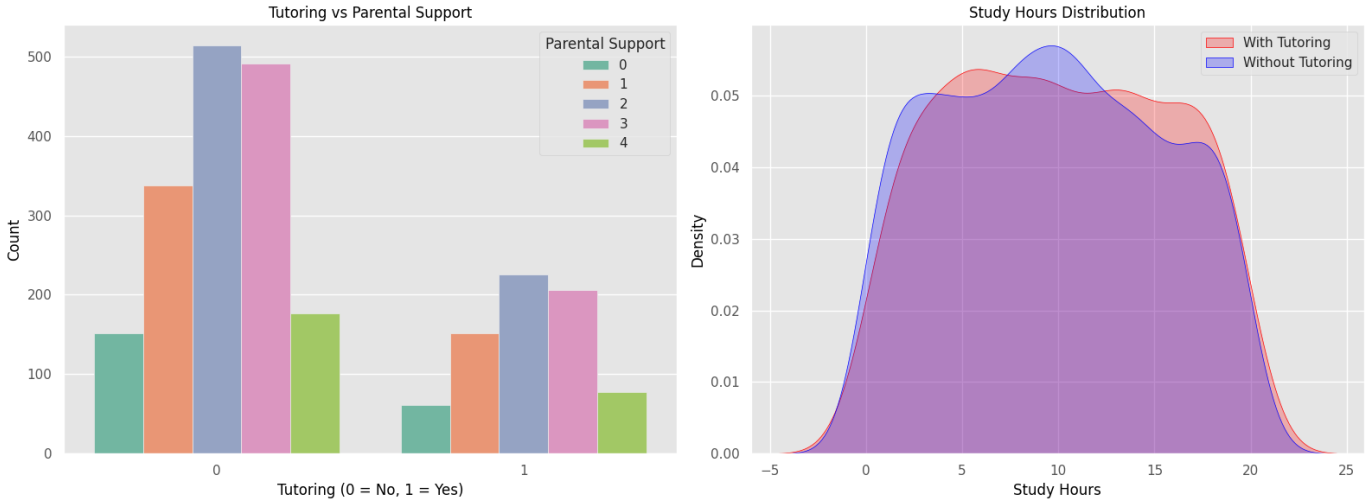


Figure 4: Study time by tutoring status and parental support.

Students who receive tutoring tend to study more weekly hours, indicating higher academic motivation as shown in Figure 4. Parental support levels, however, remain similar regardless of tutoring status.

4.2.2 Feature Engineering and Selection

A feature engineering process was used to transform the raw features of our dataset into more informative and useful features for modeling. Initially, two new features; TutoringEffect, which combined tutoring sessions with weekly study time, and ParentalInfluence, which combined parental education and parental support were introduced. Although, these features aimed to provide better insights on the relationships between external support factors and academic performance, they did not significantly improve model accuracy and instead they were potentially skewing model insights. Thus, they were removed from the final model to maintain accuracy.

To determine feature importance, a Random Forest Regressor was trained on the dataset. This process identified GPA, Absences, Study Time Weekly, and Parental Support as highly impactful features for predicting academic performance, confirming insights obtained from exploratory data analysis. To enhance model efficiency and accuracy, we removed StudentID feature due to its lack of predictive relevance.The Ethnicity feature was also removed as it introduced biases in the model despite custom weighting this feature to mitigate this issue. GradeClass was also excluded since GPA provided a

more precise measure of academic performance and the use of one-hot for GradeClass would have removed the natural order of grades. Please refer to Table 2 in the Appendix for detailed feature importance scores.

5 Methodology

5.1 Baseline Model

The baseline model for this problem begins with simple terminal prompts that ask for all relevant user information. This is processed and sent to a Random Forest Regression model, which predicts the corresponding GPA. Then, using coordinate descent, different iterations of habits and activities are tried in the goal of minimizing the difference between the GPA predicted by the Random Forest Regressor and the student's target GPA. The habits and activities that minimize this difference are returned to the user. This baseline model was developed in notebook 3.01.

5.2 Model Iteration

First classification models were briefly explored and compared to regression models. Generally, regression models offered more precision compared to classification models. This precision is useful for the optimization step of finding the best study habits. Therefore, regression was primarily explored for all future model iterations.

Tree-based models were explored (in notebook 3.11), including XGBoost and CatBoost. Both of these models were chosen because they perform quite well on tabular data. XGBoost offers high performance and accuracy, and includes regularization to prevent over-fitting. Since XGBoost is sensitive to its hyperparameters, we tuned them using Bayesian optimization. CatBoost is another popular gradient boosting algorithm that performs well with categorical features.

Various neural networks were also explored, including a multi-layer perceptron, TabNet (7), TabPFN (8), and a one-dimensional Soft-Ordering CNN (9) in notebook 3.12. An MLP was mostly used as a baseline neural network for comparison. TabNet is a neural network specialized for tabular data, and TabPFN is a prior-data fitted network that has been found to outperform many other tabular data models. Finally, the 1-dimensional soft-ordering CNN was inspired by this article (9) which outlined the model's strengths, especially when combined with TabNet.

After evaluating individual models, we saw that some models might have different strengths that could balance one another in an ensemble model. The ensemble models tried were a combination of TabNet with the 1D CNN(notebook 3.14), XGBoost with TabPFN (notebook 3.15), and a stacked ensemble model with all four neural networks, whose predictions were processed by an XGBoost model, to make the final prediction (notebook 3.16).

All of these models were trained on a CPU for less than one hour.

6 Experiments and Results

6.1 Experimental Setup

The dataset was split into a training, validation, and testing set using a 70/15/15 ratio. This split was done using the train-test-split function from Scikit-learn to ensure consistency across the experiments. Models were trained and tested throughout epochs using the training and validation set. Then, they were compared to one another on the testing set, using R^2 as the primary metric and mean-squared error and mean absolute error as secondary metrics (to better inform our understanding of the models' strengths).

6.2 Results

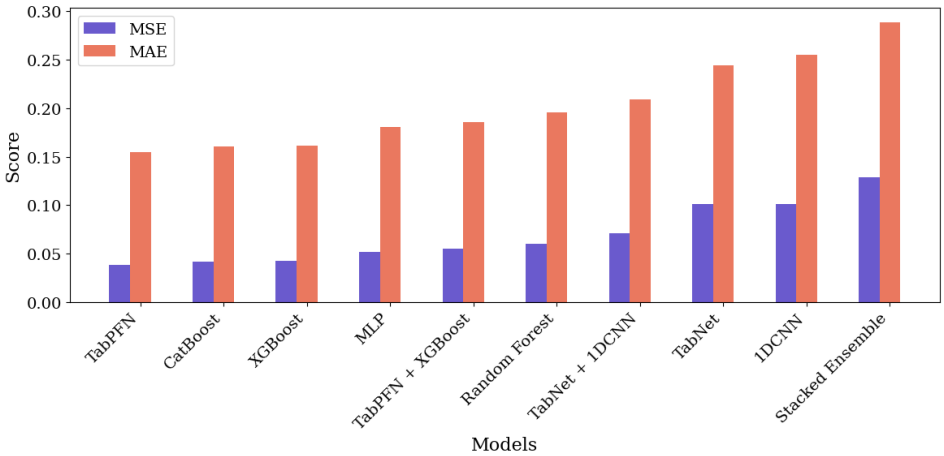
The table below displays the results from the evaluation notebook and Figure 5 displays the results of all the individual models on the testing data set.

When ensemble models were later tested, they were not able to beat the R^2 score of TabPFN. However, we hope to experiment more with ensemble models in the future.

Model	R^2 Score	MSE	MAE
TabPFN	0.952738	0.038168	0.154699
CatBoost	0.948126	0.041893	0.160363
XGBoost	0.947320	0.042544	0.161189
MLP	0.936020	0.051669	0.180589
TabPFN + XGBoost	0.931120	0.055627	0.185808
Random Forest	0.925756	0.059959	0.195237
TabNet + 1DCNN	0.880716	0.096333	0.249186
TabNet	0.874950	0.100989	0.244061
1DCNN	0.874649	0.101233	0.254664
Stacked Ensemble	0.847399	0.128495	0.288554

Table 1: Table of all R^2 scores, mean squared error and mean absolute error, for all models.

Figure 5: Bar graph comparing the loss scores, mean squared error and mean absolute error, for all models, ordered by descending R^2 values.



The habit optimizer model was trained in notebook 3.21 using both coordinate descent and bayesian optimization. Bayesian optimization (using the SkOpt library) was able to match the target GPA the closest and was much faster.

Finally, we experimented with various iterations of k-nearest neighbors in notebook 3.31 to find similar students to the user. These iterations included k-nearest neighbors without scaling, with scaling, and with weighted features. The weighted features gave the most relevant results (measured by difference in GPA and difference in other fixed user features like gender, age, and parental education).

7 Final Modeling Decisions

The final selected model is visualized in Figure 6. It uses a Graphical User Interface, powered by Gradio. User data is collected, processed and sent to two different models:

1. **Habit Recommendation System:** The pre-trained TabPFN uses the student's habits to predict their current GPA and a bayesian optimization algroithm is used to iteratively change the student's habits to minimize the difference between their desired and predicted GPA.
2. **Example Finder:** A weighted k-nearest neighbors is used to find the closest data point to the user (where GPA is substituted for desired GPA). The desired GPA is weighted the highest, followed by features that the user cannot change like gender, age, and parental education. The top 5 nearest neighbors are returned.

The results from both models are fed to the pre-trained GPT-4 model.

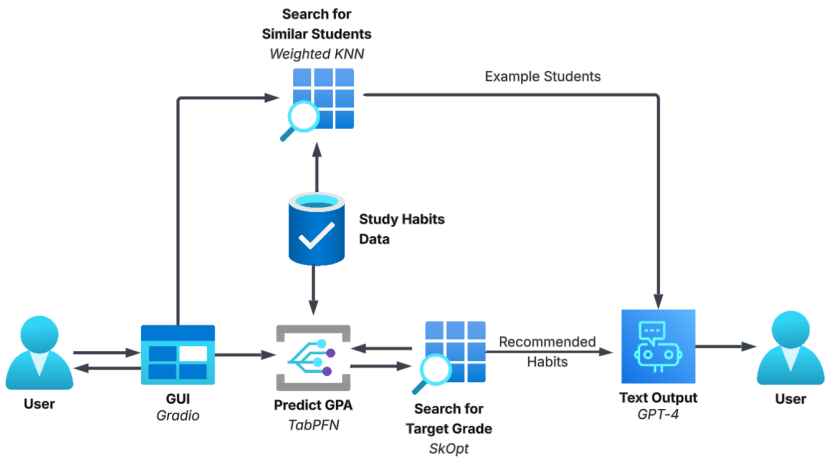


Figure 6: Final high-level architecture of SmartStudy.

7.1 GPT-4 Integration

To translate technical and numerical recommendations into actionable academic advice, we integrate OpenAI's pretrained GPT-4 model into the SmartStudy. This bridged the gap between the model's outputs and students' understanding by generating personalized, actionable feedback, what one might expect from an academic advisor. The prompt below was used to generate the advisory message:

> *"You are an academic coach. A student wants to raise their GPA to* `desired_grade`*. Please explain how these changes will help the student reach their goal. Use clear, encouraging language. Keep it under 200 words and write as bullet points."*

The GPT-4 API with a temperature of 0.7 and 'max_tokens' parameter was set to 600 was used to encourage creativity and personalization in the output, while maintaining coherence and ensuring that the response covers behavioral changes in detail. This model was used instead of older models like GPT-2 and GPT-3 because it gives more natural, clear, and thoughtful feedback. We tested multiple different prompts, including long-form paragraph responses and structured instructions. These earlier versions were informative but often too wordy or very formal in tone which might come overwhelming for some students. After refining the style and structure, we found that a bullet-point prompt with a 200-word limit was the most effective as it was concise, encouraging and easier to follow and understand. The output of our final prompt is shown below:

> *- Reducing absences: Regular class attendance is critical...*
> *- Starting tutoring: Tutoring provides personalized attention...*
> *- Increasing parental support: More involvement from parents can help...*
> *- Participating in sports: Builds time management and resilience...*
> *- Final message: Stay consistent, believe in yourself, and keep going!*

The above recommendation significantly improves the SmartStudy's output usability, clarity, and motivational. For a complete sample prompt and full GPT-4 output, refer to Appendix 11.

8 Limitations

Despite the promising results of SmartStudy, there are several limitations. Since the model is trained using synthetic data, it may not reflect real-world student behavior and also does not consider factors such as emotional well-being, socioeconomic status, and school-specific variables, which might affect students' performance. Furthermore, some recommendations generated by our model may not be actionable for all students due to limited access to resources. The feedback generated by GPT-4 also depends on the given prompt and may vary in tone or clarity. Lastly, the system does not adapt to real-time user behavior, which limits the long-term personalization. Future improvements should address these issues to enhance fairness, generalizability, and real-world effectiveness.

9 Conclusion and Future Work

The results of this study demonstrate the potential of SmartStudy, a machine learning tool, to help students improve their grades by providing personalized study and lifestyle recommendations. By examining factors like absences, parental help, and study hours, the tool finds ways to boost GPA based on trends explored in the dataset. Among the models tested, TabPFN delivered the most accurate predictions (R^2 value of 0.953), outperforming alternatives like Random Forest and XGBoost.

To increase accessibility and interactivity, we plan to develop a web-based demo of SmartStudy where students can input their study habits, extracurricular activities, and academic goals to receive customized recommendations. We will also gather user feedback through surveys and engagement tracking to assess usability and effectiveness. Metrics such as user satisfaction, interaction frequency, and qualitative responses will guide refinements to the tool's logic and user interface. Looking ahead we aim to make this tool more adaptive through real-time feedback loops, allowing it to continuously learn from user behavior. Additionally, integrating SmartStudy with educational and productivity tools such as Notion and Google Calendar will allow students to seamlessly incorporate recommendations into their daily routines.

SmartStudy has the potential to support students who may not have access to traditional academic resources such as tutoring or mentorship, by offering low-barrier, personalized guidance. Ensuring fairness and ethical use is critical in our model, therefore, as part of our future work, we will explore fairness-aware modeling to offer students more control through explainability and override options.

SmartStudy demonstrates that with the right tools, machine learning can empower students to study smarter and take ownership of their academic journey.

10 Acknowledgments

The authors thank ECE324's course instructors and teaching team at the University of Toronto for their guidance throughout this project. We also acknowledge the creators of the Students Performance Dataset and OpenAI for providing access to GPT-4, which enabled the recommendation component of our system.

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11 Appendices

Appendix A: Feature Importance Scores

The table below shows the Feature Importance Scores that were obtained using Random Forest Regressor

Table 2: Feature Importance Scores

Rank	Feature	Importance
1	Absences	0.741183
2	GradeClass	0.157313
3	StudentID	0.030647
4	StudyTimeWeekly	0.028357
5	ParentalSupport	0.016383
6	Tutoring	0.005534
7	Sports	0.003650
8	Extracurricular	0.003552
9	ParentalEducation	0.003548
10	Age	0.003360
11	Ethnicity	0.002587
12	Music	0.001487
13	Gender	0.001342
14	Volunteering	0.001057

Appendix B: Full GPT-4 Prompt and Output Example

Final Prompt:

> *You are an academic coach. A student wants to raise their GPA to* `desired_grade`*. Please explain how these changes will help the student reach their goal. Use clear, encouraging language. Keep it under 200 words and write as bullet points.*

This prompt was accompanied by structured information about the student's current and optimized habits, including:

- Current Study Time: 10 hours/week → Optimized: 10 hours/week
- Absences: 3 → Optimized: 0
- Extracurricular: No → Optimized: Yes
- Music: No → Optimized: No
- Volunteering: No → Optimized: No
- Parental Support: 1 → Optimized: 4
- Tutoring: No → Optimized: Yes
- Sport: No → Optimized: Yes

- Desired GPA: 3.8

Sample Recommendation Output:

- Reducing absences: Regular class attendance is critical to understanding the material, participating in discussions, and keeping up with assignments. This change will ensure you don't miss out on any learning opportunities and directly support GPA improvement.
- Starting tutoring: Tutoring provides personalized attention and helps with challenging subjects. It's a great step toward mastering the material and improving your grades.
- Increasing parental support: Greater involvement from your parents can offer additional motivation, structure, and encouragement. They can assist in organizing your study schedule and keeping you focused.
- Participating in sports: Engaging in extracurriculars like sports builds time management skills, resilience, and a growth mindset—all of which positively impact academic performance. It also helps maintain a healthy study-life balance.

Remember, every small step counts. Stay consistent, believe in yourself, and you're well on your way to achieving your GPA goal.